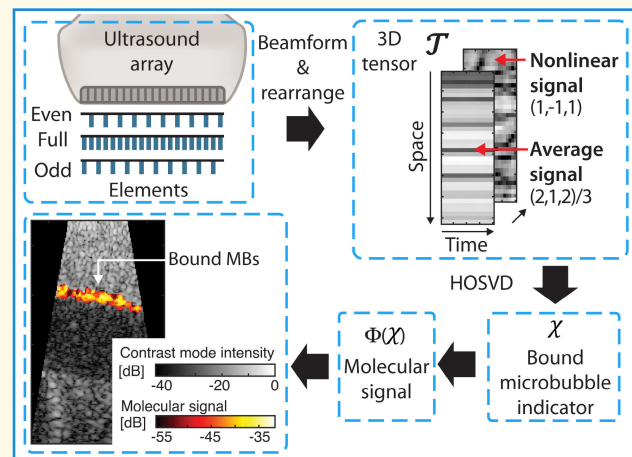


Nondestructive Ultrasound Molecular Imaging With Higher Order Singular Value Decomposition

Gonzalo Collado-Lara¹, Member, IEEE, Geraldi Wahyulaksana², Member, IEEE, Hendrik J. Vos¹, Member, IEEE, and Klazina Kooiman¹, Senior Member, IEEE

Abstract—Ultrasound molecular imaging (UMI) uses targeted microbubbles (MBs) to detect disease-associated biomarkers. For UMI, distinguishing the acoustic signals produced by bound MBs from those by free MBs and tissue is critical. Currently, the main approach, known as differential targeted enhancement (DTE), is time-intensive and requires MB destruction. Here, we introduce a novel, rapid, and nondestructive UMI technique utilizing higher order singular value decomposition (HOSVD). HOSVD decomposes the signals of an acoustic contrast sequence, separating them owing to their nonlinear content and temporal coherence. The nonlinear separation enables distinction between tissue and MBs, while the temporal separation enables distinction between free and bound MBs. From the HOSVD output, we defined a bound MB indicator χ , which indicates the presence of bound MBs. In our in vitro experiments, χ was lower for free MBs and tissue (0.04 ± 0.03) compared to bound MBs (0.31 ± 0.11 without free MBs, decreasing with concentration down to 0.11 ± 0.07 at 20×10^3 free MBs/mL). In addition, the molecular signal determined from χ correlated well with a DTE ground truth acquisition. The method was compared to other nondestructive techniques such as low-pass filtering and normalized singular spectrum area, demonstrating an average molecular signal enhancement of 12 dB. Furthermore, when used as a binary classifier, our method achieved a detection of up to $1.81 \times$ more true positives while reducing false positives by up to $1.78 \times$. These findings suggest that HOSVD could pave the way to rapid, nondestructive UMI.

Index Terms—Contrast-enhanced ultrasound, molecular signal, nondestructive, targeted microbubbles (MBs), ultrasound contrast agents, ultrasound molecular imaging (UMI).



I. INTRODUCTION

MOLECULAR imaging seeks a noninvasive diagnosis by detecting the expression of certain biomarkers [1].

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Ultrasound molecular imaging (UMI) uses microbubbles (MBs), 1–10 μm in diameter [2] that, upon intravenous injection, disperse throughout the vasculature. MBs decorated with ligands [3] can target and bind to biomarkers related to, among others, inflammation, cancer, thrombus, and bacterial infection [4], [5], [6], [7], [8], [9]. Monitoring the MB binding using ultrasound imaging enables a noninvasive diagnosis [10]. For this purpose, separating the acoustic signals from bound MBs (i.e., molecular signal) from the signals produced by surrounding tissue and free unbound MBs is critical [11].

The most popular technique used to isolate the molecular signal is known as differential targeted enhancement (DTE) [12]. In DTE, after waiting several minutes for MBs to bind, all MBs are burst. Perfusion then replenishes free MBs over a few minutes, yet no further MB binding is expected. Thus, the difference between pre- and post-burst images reveals the molecular signal. Although effective, DTE is time-costly and the MB destruction may induce undesired biological

Highlights

- We propose higher order singular value decomposition for rapid and nondestructive ultrasound molecular imaging based on the nonlinear behavior and spatiotemporal patterns of bound microbubbles.
- Our method provided higher molecular signal intensity (an average of 12 dB) and specificity (up to $1.8\times$) than other nondestructive methods.
- The proposed method holds promise for accurate, real-time molecular imaging, eliminating the waiting steps necessary in the destructive gold standard, i.e., differential targeted enhancement.

effects [13]. Instead, a rapid, nondestructive UMI technique is desired [14].

Other UMI techniques than DTE often use two physical mechanisms: the different temporal patterns of bound and free MB signals over multiple frames [11], and/or the higher nonlinear response of MBs compared to tissue [15]. Due to circulation, free MBs move rapidly, while the motion of bound MBs and tissue is generally slower. Low-pass filters (LPFs) [16], [17] and singular value decomposition (SVD) [18] are commonly used to separate fast from slow motion. SVD decomposes the acoustic acquisition into an orthogonal basis, where temporally coherent signals are grouped together. Normally, lower velocities such as tissue motion and slow-moving MBs occupy the lower ranks, while higher velocities (i.e., flowing MBs) occupy the higher ranks. Filtering is achieved by the selection of the desired ranks [19].

Nonetheless, temporal analysis alone fails to separate bound MBs from tissue due to their similar temporal patterns. Instead, contrast-pulsing sequencing (CPS) is used to isolate the nonlinear response of MBs. CPS combines the acquisition of multiple pulses with equal frequency but varying amplitude or polarity to discard the linear content while retaining the nonlinear signals [20]. Nonetheless, **tissue response is not completely linear [21], and strong tissue reflections could be miscategorized as bound MBs [22]**, especially when nonlinear propagation accumulates in the transmitted wave [23].

To overcome the previous limitations, other methods have been proposed. Mauldin et al. [24] combined block-wise SVD (i.e., to subgroups of pixels) with long acoustic radiation force (ARF) pulses [25]. They defined a variable called normalized singular spectrum area (NSSA), which was related to the temporal changes within a block. ARF pulses were used to induce a decorrelation in the signal of bound MBs, increasing their NSSA compared to tissue [25]. The molecular signal was retained by filtering out low-NSSA blocks with tissue and high-NSSA blocks with free MBs. However, the NSSA of bound MBs greatly changed under different conditions, often overlapping with the NSSA of free MBs [26]. Furthermore, ARF-induced motion caused MBs to temporarily adhere to tissue [27], potentially leading to false positives (FPs). Another proposed method involved deep learning network trained with DTE recordings [28]. However, this technique is yet unable to separate bound from free MBs, and generalization of the deep learning model can be challenging and requires large training datasets. In a different approach, Turco et al. [29] proposed separation of bound and free MBs based on the MB

concentration kinetics after injection. Areas of high MB retention are expected to contain bound MBs. While promising, the model requires histological validation relating the kinetics with the biomarker expression. Thus, rapid, nondestructive UMI is desirable, yet no technique has replaced DTE due to their individual limitations.

Here, we propose a novel approach based on the block-wise application of the multilinear generalization of SVD, i.e., higher order SVD (HOSVD). HOSVD was first introduced in the context of ultrasound imaging for noncontrast blood flow imaging, where it outperformed conventional SVD [30], [31]. When applied to a contrast acquisition, HOSVD decomposes the signals' temporal patterns and their nonlinear content [32], which we expect to enable the separation of bound MBs from free MBs and tissue simultaneously. For this purpose, we derived a novel metric that represents the presence of bound MBs and the bound MB indicator χ and validated its performance using in silico simulations. Then, we used χ to determine the molecular signal of MBs bound to a tissue-mimicking phantom in vitro with increasing concentrations of flowing MBs, which was validated against DTE and compared to LPF and NSSA.

II. THEORY

UMI requires the separation of bound MB signals from free MB and tissue signals. Conventional SVD applied to U.S. recordings [18] decomposes the signals into multiple ranks, where each rank represents unique spatiotemporally coherent signals. The temporal coherence of the signals from bound and free MB is low due to their different motion patterns, while tissue and bound MBs move together, producing temporally coherent signals [25]. Thus, free and bound MBs signals are represented by different ranks [33], yet bound MBs and tissue signals are grouped into the same ranks [34], so tissue and bound MBs signals cannot be separated by SVD rank selection. Instead, we propose to solve this limitation using HOSVD to exploit the higher nonlinear response of MBs compared to tissue.

HOSVD is a multilinear SVD generalization that enables the decomposition of n -order tensors [35], thereby allowing the inclusion of multiple pulses of a contrast sequence along a third dimension. HOSVD decomposes the signals in three dimensions: spatial, temporal, and third dimension where the linear and nonlinear content of the signals are separated into different ranks [32]. We hypothesize that the signal of bound MBs will be represented together with tissue in the

lower temporal ranks. These may be separated from the free MB signals contained in the higher temporal ranks by rank selection, as in conventional SVD [30], [31]. Furthermore, the nonlinear signal of bound MB may be separated from the linear signal of tissue by rank selection along the third dimension.

A. HOSVD Implementation

The implementation of HOSVD is depicted in Fig. 1. Using a three-pulse ($N_p = 3$) amplitude modulation (AM) sequence, with even, all, and odd elements [36], N_t frames are acquired. The received signals are then beamformed to generate in-phase/quadrature (IQ) data. A block of $N_x \times N_z$ pixels is selected, and the IQ data are reshaped into three matrices where the columns represent the temporal values of each pixel. The matrices are combined twice ($N_c = 2$) to obtain the average (+2/3, +1/3, +2/3) and nonlinear signal (+1, -1, +1). This preprocessing step minimizes artifacts, as will be discussed later in Section V. The resulting signals are concatenated along a third dimension to form a tensor $\mathbf{T}$ with a spatial, temporal, and pulsing dimension. The tensor is decomposed by means of HOSVD

$$\mathbf{T} = \sum_i^{N_x N_z} \sum_j^{N_t} \sum_k^{N_c} \mathbf{S}_{i,j,k} \mathbf{U}_i \mathbf{V}_j \mathbf{W}_k \quad (1)$$

in which $\mathbf{S}$ is the HOSVD core tensor of size $N_x N_z \times N_t \times N_c$ [35], and $\mathbf{U}$, $\mathbf{V}$, and $\mathbf{W}$ are the orthonormal spatial, temporal, and pulsing basis, respectively.

B. Bound MB Indicator

Based on our previous hypotheses, our aim is to detect the bound MBs by finding nonlinear signals that are temporally coherent with tissue signals. For this, a bound MB indicator is proposed, here defined for the absence of tissue motion

$$\chi = \sqrt{\frac{\sum_i^{N_x N_z} (\mathbf{S}_{i,1,nl} \mathbf{S}_{i,1,nl}^*)}{\sum_i^{N_x N_z} \sum_j^{N_t} \sum_k^{N_c} (\mathbf{S}_{i,j,k} \mathbf{S}_{i,j,k}^*)}} \quad (2)$$

where 1 indicates the first temporal rank, nl is the index referring to the nonlinear rank, and $\mathbf{S}^*$ is the complex conjugate of the tensor core. The variable χ represents the fraction of nonlinear content of all the spatial ranks and the first temporal rank over the total energy of all ranks, and its definition was based on three hypotheses. First, in the absence of tissue motion, tissue signals will be grouped in the first temporal rank. Second, the temporal coherence of bound MB and tissue signals is high, so the signal of bound MBs will also be grouped in the first temporal rank, increasing its nonlinear content. Third, the incoherent signal of free MBs will spread across multiple ranks. Therefore, the numerator of (2) will increase in the presence of bound MBs, while the denominator will increase in the presence of free MBs. Accordingly, χ is expected to be higher for blocks containing bound MBs than for blocks containing tissue and/or free MBs.

We defined the MB indicator as the nonlinear to total energy fraction to mitigate artifacts introduced by nonlinear

propagation. This artifact arises when the imaging wave is distorted as it propagates through tissue and/or MBs [37]. Thereafter, reflected signals will have some nonlinear content. Consequently, just as for bound MBs, the nonlinear content of the signal reflected by stationary reflectors will be expressed in the first temporal rank. However, as bound MBs are strong sources of nonlinearity, we expect that the fraction of nonlinear to linear energy scattered by the MBs will be much higher than the fraction of nonlinear to linear energy contained in the imaging wave after nonlinear propagation.

C. Molecular Signal

While χ relates to the presence of bound MBs in a given block, it does not preserve information of the signal intensity in that block. We therefore defined the molecular signal as

$$s_{\text{HOSVD}} = s_{\text{AM}} \cdot \Phi((\chi - \mu_\chi)/\sigma_\chi) \quad (3)$$

where s_{AM} is the AM signal, averaged in slow-time over all frames, and Φ is the cumulative density function (cdf) of a normal distribution with mean μ_χ and standard deviation σ_χ . Consequently, $\Phi(\chi)$ will filter out blocks with a low χ while retaining blocks with a high χ .

III. MATERIALS AND METHODS

A. In Silico Experiment

An in silico experiment was used to validate χ for bound MB detection. We simulated an AM acquisition using the Proteus platform [38], which integrates k -Wave as an acoustic solver [39] and a modified Rayleigh–Plesset equation to account for the nonlinear behavior of MBs [15]. The simulation domain (see Fig. 2), which scattered ultrasound due to impedance inhomogeneities, included two slanted tubes that intersected at the center of the domain. One tube contained bound MBs and the other free MBs, both at a concentration of 35 MBs/mm². MBs had all equal properties, but bound MBs were completely stationary, whereas free MBs traveled away from the transducer at 5 cm/s. Other simulation parameters are detailed in Supplementary Table 1.

B. In Vitro Experiment Materials

In-house biotinylated MBs were produced by tip sonication of a lipid mixture of DSPC (90 mol%, kindly provided by Lipoid, Ludwigshafen, DE, Germany), DSPE-PEG5000 (8.9 mol%, Nanosoft Polymers, Winston-Salem, NC, USA), and DSPE-PEG5000-biotin (1.1 mol%, Nanosoft Polymers) in a perfluorobutane (C₄F₁₀, F2 Chemicals, Preston, U.K.) atmosphere, following the indirect method described in [40]. The size distribution of the washed MBs was measured using a Coulter Counter Multisizer 3 (Fig. S1, 50- μ m aperture tube, Beckman Coulter, Mijdrecht, The Netherlands).

Tissue-mimicking phantoms were made of milli-Q water, agar (2% w/v, Thermo Fisher Scientific, Waltham, MA, USA), and silicon carbide particles (4% w/v, ~400 mesh, Sigma Aldrich, St. Louis, MI, USA). After boiling, the solution was poured into molds, embedding a nylon inclusion (4.9 $\times$ 1.5 mm cross section) to act as an acoustic reflector. Then, one surface of the phantoms was coated with

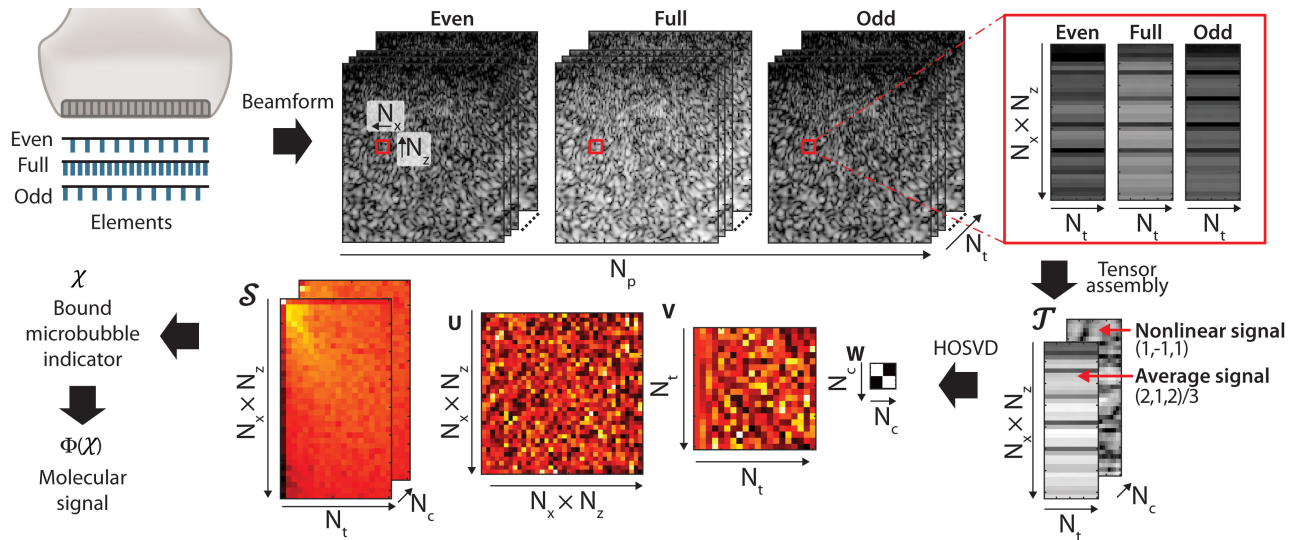


Fig. 1. Implementation of HOSVD for UMI. The beamformed data of an AM sequence are analyzed block-wise. The block data are reshaped into matrices whose rows represent the temporal evolution of individual pixels. The matrices are then linearly combined to obtain the average and the nonlinear signal and concatenated to form a 3-D tensor $\mathbf{T}$. The HOSVD of $\mathbf{T}$ yields the orthonormal basis $\mathbf{U}$, $\mathbf{V}$, $\mathbf{W}$, and the tensor core $\mathbf{S}$, which is used to calculate the bound MB indicator χ . Finally, χ is used to determine the molecular signal.

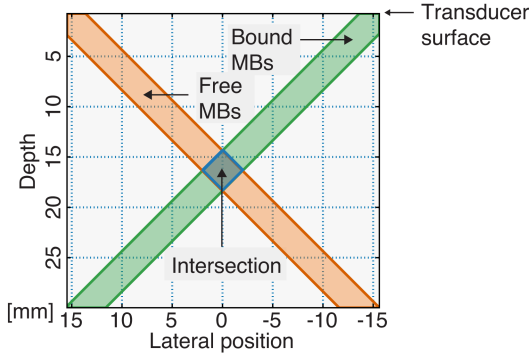


Fig. 2. In silico domain. A tube containing free MBs (orange) and a tube containing bound MBs (green) intersect at the center of the domain.

Avidin Texas Red¹ conjugate (Invitrogen, Thermo Fisher Scientific) overnight at a concentration of 50 $\mu\text{g}/\text{mL}$ [41] and subsequently incubated with 5% bovine serum albumin (BSA, Sigma Aldrich) for 1 h at room temperature to minimize nonspecific binding. Afterward, MBs were bound to the functionalized surface by allowing them to float against the treated surface in dulbecco's phosphate-buffered saline (DPBS) Ca^+Mg^+ for 10 min (phantoms with bound MBs). The bound MBs at ten different locations were counted using confocal microscopy (Nikon A1R+, Nikon Instruments, Tokyo, JP) Z-stacks with 2- μm steps at a resolution of 0.62 $\mu\text{m}/\text{pixel}$ using a 20 $\times$ water immersion objective (Cfi Apo LWD, Nikon Instruments). The bound MB concentration was $274 \pm 150 \text{ MBs}/\text{mm}^2$ (mean $\pm$ standard deviation, Fig. S2). As a control, phantoms were only incubated with 5% BSA (phantoms without bound MBs).

C. Ultrasound Acquisitions

Fig. 3 depicts the in vitro acoustic setup. The phantoms were attached to a motorized stage and immersed in a closed

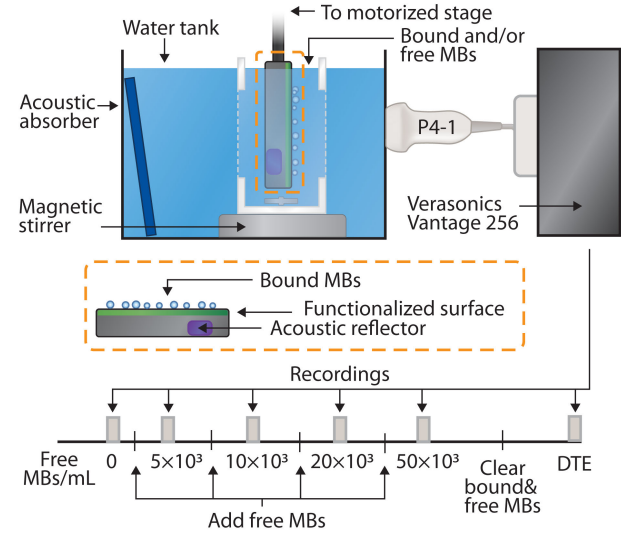


Fig. 3. Setup used for the acoustic recordings. Ultrasound recordings were obtained at several concentrations of free MBs. After the bound and free MBs were removed, a DTE ground truth recording was obtained.

container [poly methyl methacrylate (PMMA)] with acoustically transparent windows (polyester, 20 μm thickness). The container was located in a water tank, together with an acoustic absorber (Aptflex F28, Precision Acoustics, Dorchester, U.K.) that minimized multiple reflections. The container was filled with DPBS, which was continually stirred. Ultrasound acquisitions were recorded using a phased P4-1 array connected to a Vantage 256 system (Verasonics, Kirkland, WA, USA). The acquisitions consisted of an AM sequence of three pulses (even, all, and odd elements), with 300 μs between pulses and a pulse repetition rate of 1000 Hz for 1 s. Transmissions consisted of 15 $^\circ$ diverging waves, centered at 2 MHz, 1.5 cycles length, and 30-kPa maximum peak negative pressure (PNP), measured with a 0.2-mm needle hydrophone (Precision Acoustics).

¹Trademarked.

Six conditions were recorded for each phantom: 1) without free MBs; 2)–5) various concentrations of free MBs (i.e., 5×10^3 , 10×10^3 , 20×10^3 , and 50×10^3 , MBs/mL); and 6) after clearing the free MBs and destroying the bound MBs with acoustic bursts (100 cycles, 2.1-MPa PNP). The MB concentrations were chosen to cover a range of realistic clinical concentrations. Two phantoms with and two without bound MBs were imaged, and three recordings were acquired for each phantom at each condition, moving the phantom 5 mm transversally to the transducer between recordings, resulting in six recordings per condition.

D. Data Analysis

Data analysis was done with MATLAB 2022a (The Mathworks, Natick, MA, USA).

1) *In Silico Experiment*: The resulting RF signals were beamformed using a delay-and-sum algorithm and a pixel size of $\lambda/4$, where λ is the transmitted wavelength. HOSVD (Tensorlab [42]) was applied to 5×5 pixel blocks, and χ was calculated using (2). The block moved one pixel laterally or vertically, so adjacent blocks had a 20-pixel overlap. Importantly, the HOSVD tensor core was nondiagonal (see Fig. 1), and thus, when referring to the energy of a temporal rank, we imply the sum of all spatial ranks.

2) *In Vitro Recordings*: The acoustic recordings were beamformed using delay-and-sum with a pixel size of $\lambda/4$. To calculate χ , HOSVD was applied to 5×5 pixel blocks with a 20-pixel overlap, as in the in silico experiment. Subsequently, the molecular signal was calculated using (3), where μ_χ and σ_χ were optimized to maximize the filter's performance based on the receiver operating characteristic (ROC) curves, as described under Section III-D3. For each phantom, we identified various regions of interest (ROIs): free MBs, surface (with or without bound MBs), phantom, and reflector. The average molecular signal in each ROI of the phantoms with bound MBs was calculated, obtaining the inter-recording error from the standard deviation between all six acquisitions at equal free MB concentrations. The minimum contrast was defined as the intensity difference between the bound MBs and the ROI with the highest intensity among the other regions.

3) *Comparison With Other Techniques*: The difference in nonlinear signal s_{AM} between the acquisition with only bound MBs and the acquisition clear of all MBs were used as in DTE to produce a molecular signal ground truth. This signal was qualitatively compared to the molecular signal obtained with the proposed method. The performance of HOSVD as a molecular signal filter was compared to LPF and NSSA without ARF pulses. To generate the LPF molecular signal s_{LPF} , the nonlinear signal was filtered using a zero-phase Butterworth low-pass filter (4th-order, 0.5-Hz cutoff frequency). NSSA was calculated in [24, eq. (6)] Click or tap here to enter text., using a 5×5 pixel block-wise SVD transform of the full aperture transmission. The NSSA molecular signal was calculated as

$$s_{NSSA} = s_{AM} \cdot \phi((NSSA - \mu_{NSSA})/\sigma_{NSSA}) \quad (4)$$

where ϕ represents the probability density function of a normal distribution of mean μ_{NSSA} and standard deviation σ_{NSSA} . These parameters were optimized for HOSVD.

The ROC of the filters was also studied. For this, all pixels of the ROI with bound MBs were categorized as positives (P), and all pixels in other ROIs as negatives (N). For each molecular image, pixels with intensities above a given threshold were predicted positive and otherwise predicted negative. These predictions were true positives (TPs) or true negatives (TNs) when they matched the actual condition, and FPs and false negatives (FNs) otherwise. We then calculated the optimal intensity threshold as the one that maximized the filter accuracy ($ACC = [TP + TN]/[P + N]$). For that threshold, the TP rate (TPR) ($TPR = TP/P$) and the false discovery rate ($ACC = FP/[FP + TP]$) were calculated. An algorithm was used to find μ_χ , σ_χ , μ_{NSSA} , and σ_{NSSA} that maximized TPR for each filter.

IV. RESULTS

A. In Silico Proof of Concept

Fig. 4(a) and (b) shows the linear and nonlinear content of a sample frame, respectively. The linear content [see Fig. 4(a)] backscattered by tissue and MBs was similar, whereas the nonlinear content [see Fig. 4(b)] of MBs was substantially higher. The subplots in Fig. 4(a) and (b) depict the spatiotemporal patterns of four representative blocks: bound MBs, tissue, free MBs, and the intersection between bound and free MBs. The linear and nonlinear temporal patterns differed; the linear content of bound MBs and tissue barely changed over time, whereas clear variations of the nonlinear content were observed. For the free MBs and intersection, clear changes were present in both linear and nonlinear content. Additionally, the overall nonlinear content of the intersection was higher than the other blocks, owing to the presence of both bound and free MBs. The implications of these observations on the distribution of linear and nonlinear content along the HOSVD temporal ranks are illustrated in Fig. 4(c). The linear content of the first temporal rank, primarily generated by static tissue, was similar across all four blocks. However, the nonlinear content of the first temporal rank was higher for the blocks containing bound MBs (i.e., bound MBs and intersection) than the blocks that did not (i.e., free MBs and tissue). Across higher temporal ranks, both for linear and nonlinear content, the blocks containing free MBs (i.e., free MBs and intersection) had higher energy than the blocks that did not (bound MBs and tissue). These findings support the hypothesis used to define χ .

Fig. 4(d) depicts the components of (2). Compared to the tissue block, the presence of bound MBs increased the numerator, while the presence of free MBs increased the denominator. Consequently, the bound MB indicator χ [see Fig. 4(e)] was the highest for the bound MBs block (0.08) and was higher for the intersection block (0.05), which contained both bound and free MBs, than for tissue (0.01) and free MBs (0.02) blocks. This trend held across the whole simulation domain [see Fig. 5(a)], as χ values were higher for the bound MBs (mean $\pm$ standard deviation, 0.046 ± 0.025) and intersection (0.040 ± 0.0189) compared to free MBs (0.012 ± 0.007) and tissue (0.012 ± 0.008).

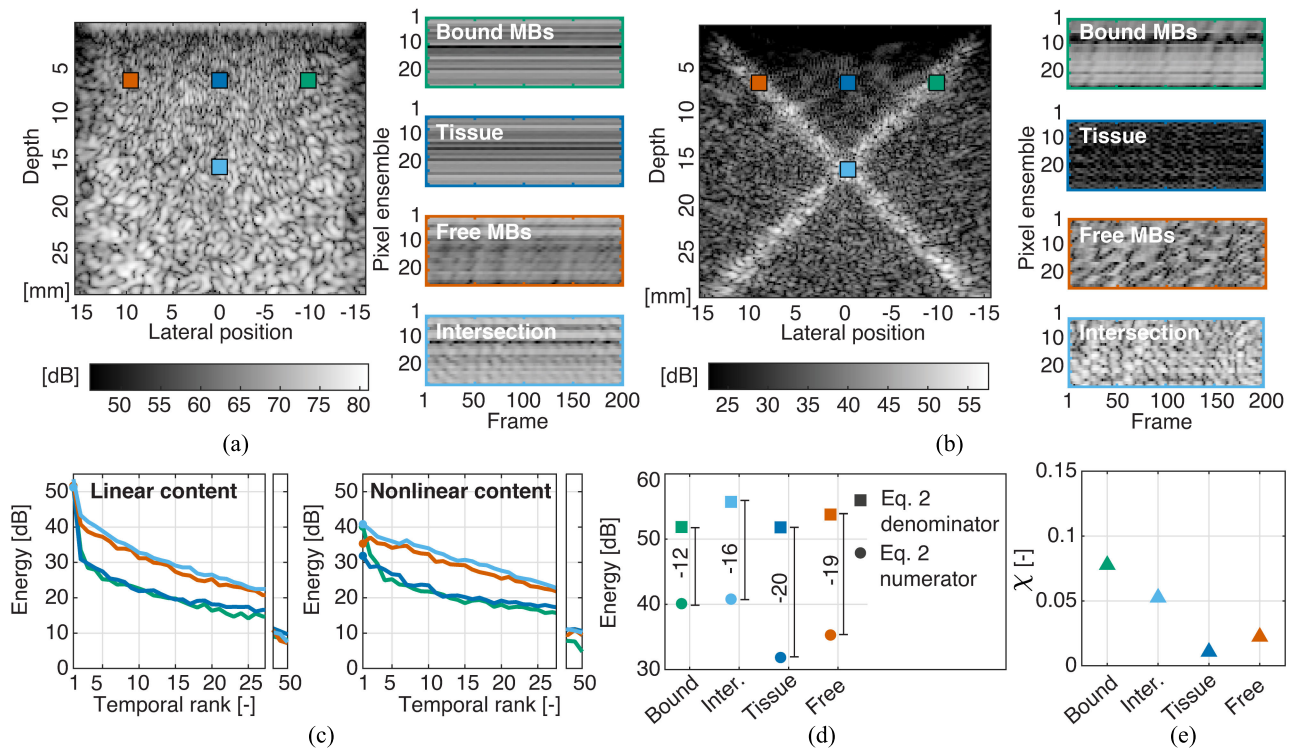


Fig. 4. In silico proof of concept. Sample frames of (a) linear and (b) nonlinear content. The subplots represent the temporal evolution of the signal in representative blocks of bound MBs (green), tissue (dark blue), free MBs (orange), and intersection of bound and free MBs (light blue). (c) Linear and nonlinear energy of the temporal ranks. (d) Values of the numerator and denominator of (2), with their difference in dB. (e) χ values of the four sample blocks. (Bound = Bound MBs, Inter. = Intersection, and Free = Free MBs).

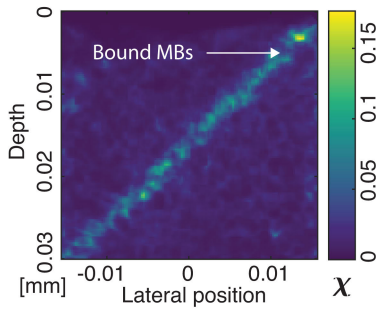


Fig. 5. χ distribution in the simulation domain calculated with HOSVD.

B. In Vitro Experiment

Fig. 6(a) shows the sample frames of the AM acquisitions for increasing concentrations of free MBs, and Fig. 6(b) shows the multiple ROIs in the acoustic recordings. Without free MBs, the backscattered intensity from the surface was higher for the phantom with bound MBs (top row) than for the phantom without bound MBs (bottom row), attributed to the presence of bound MBs in the former. In later acquisitions, as the concentration of free MBs increased, so did the acoustic attenuation, consequently decreasing the backscattered intensity with increasing depth.

The average value of χ in each ROI as a function of free MB concentration is shown in Fig. 6(c). In the phantom with bound MBs, χ was the highest at the surface with bound MBs. This value (up to 0.31 ± 0.11 at 0 free MBs/mL),

decreased with increasing concentration of free MBs, until it overlapped with the other ROIs at 50×10^3 free MBs/mL. In the remaining ROIs of both phantoms with and without bound MBs, χ ranged from 0.02 to 0.07, including the surface of the phantoms without bound MBs.

Molecular signal images were generated by combining the AM frames and the χ values using (3) [see Fig. 7(a)] for the same acquisitions as in Fig. 6. In the phantom with bound MBs, the molecular signal generated by the bound MBs was visible up to a concentration of 20×10^3 free MBs/mL (25 dB dynamic range). In the phantom without bound MBs, no strong molecular signal was present at the surface. These findings were consistent with the DTE acquisitions [see Fig. 7(b)], which showed a strong MB signal at the surface of the phantom with bound MBs but not the phantom without bound MBs.

Some molecular signal clusters were observed in both phantoms with the proposed HOSVD method at 0 and 5×10^3 free MBs/mL, which were also present in the DTE [white arrows in Fig. 7(b)]. These signals were unlikely linked to bound MBs, as the phantom without bound MBs had not been in contact with MBs in the first acquisition. Thus, due to the unknown nature of these signals, they were excluded from all ROIs and subsequent analysis.

Fig. 8 shows the average molecular signal intensity of the different ROIs for the phantoms with bound MBs processed with the HOSVD method as a function of free MBs concentration, normalized to the minimum intensity. Up to 20×10^3 MBs/mL, the bound MBs intensity was higher than

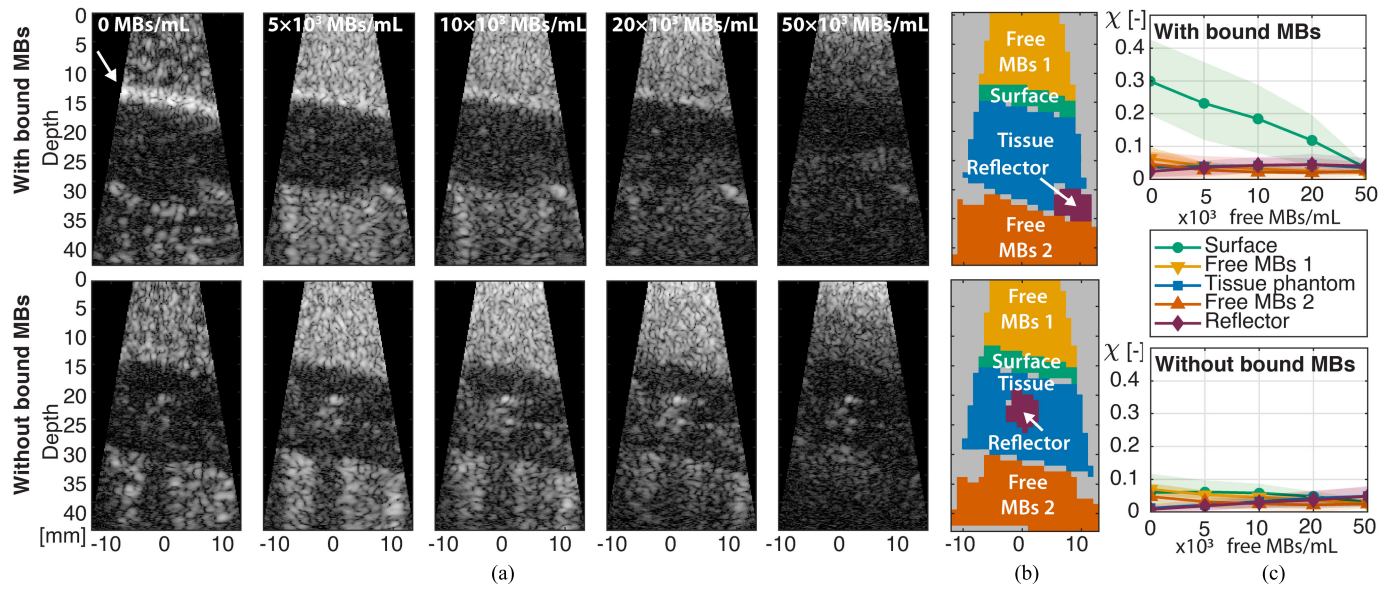


Fig. 6. In vitro validation of χ . (a) Sample AM frames with increasing concentration of free MBs (left to right), with (top) and without (bottom) MBs bound to the surface. The white arrow indicates the location of the bound MBs. (b) Defined ROIs. The sections in gray were not included. (c) Average χ values within each ROI. The shaded region represents the intra-acquisition standard deviation ($n = 6$).

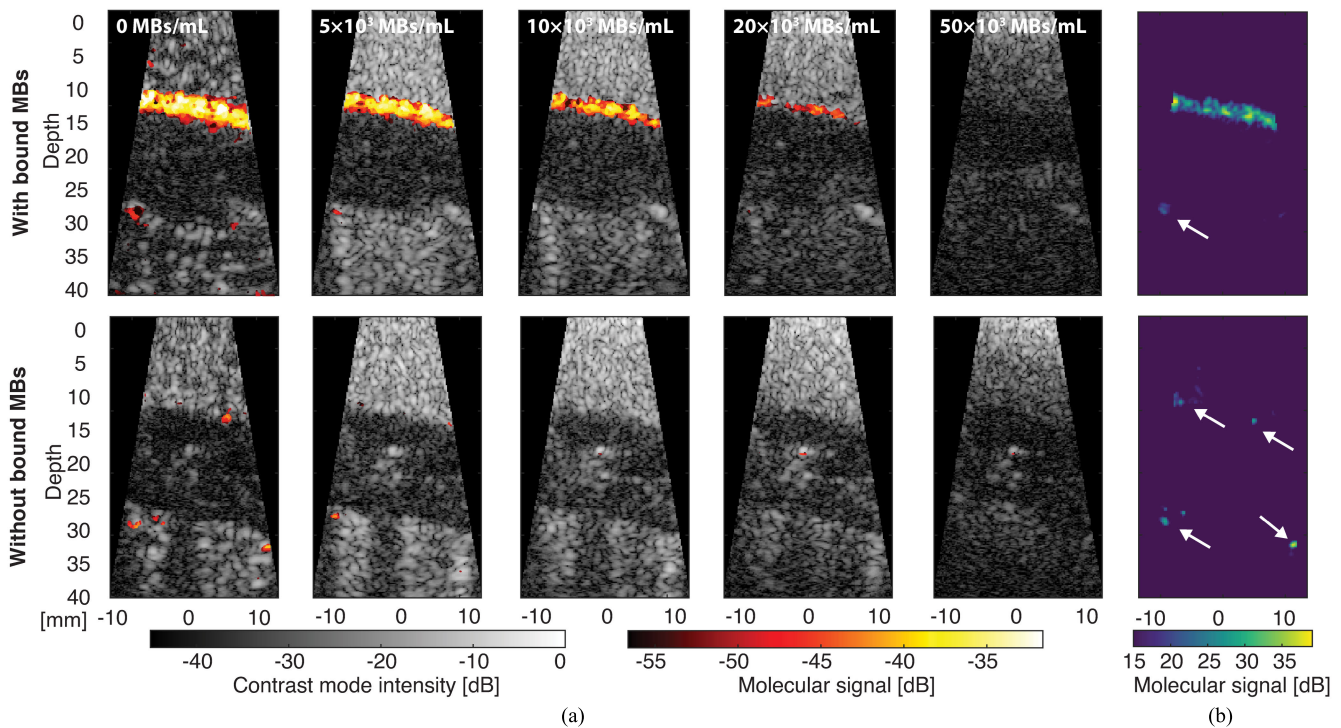


Fig. 7. Experimental UMI with HOSVD. (a) Molecular signal (in yellow-red) overlaid on an AM frame (gray) obtained using the proposed HOSVD method, for a phantom with (top) and without (bottom) bound MBs, and increasing concentrations of free MBs (left to right). (b) Ground truth obtained with DTE (in blue-yellow). The white arrows indicate signals unlikely due to bound MBs.

all other ROIs, with a value of 58.2 dB at 0 free MBs/mL, down to 44.7 dB at 20×10^3 MBs/mL. At 50×10^3 MBs/mL, the bound MBs intensity quickly fell to 6 dB. The limiting ROI (i.e., the one with the highest intensity besides the bound MBs) was the free MBs 1 up to 10×10^3 MBs/mL, and the reflector at higher concentrations. At 50×10^3 free MBs/mL, the intensity of the free MBs 1, tissue, and reflector exceeded the intensity of bound MBs.

C. Comparison With Other Techniques

The molecular signal obtained at 10×10^3 free MBs/mL using the proposed HOSVD method in comparison to the two other real-time techniques LPF and NSSA is shown in Fig. 9. The molecular signal was normalized using the maximum molecular intensity within each acquisition. HOSVD retained only the molecular signal from bound MBs, whereas LPF and NSSA exhibited leakage from free MBs and the

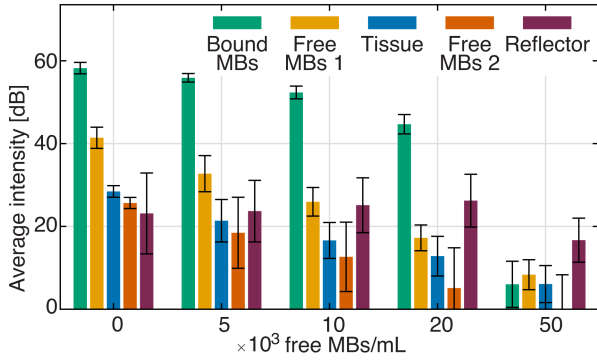


Fig. 8. Average intensity in the multiple ROIs after filtering with the proposed HOSVD method. The error bars represent the inter-recording standard deviation ($n = 6$).

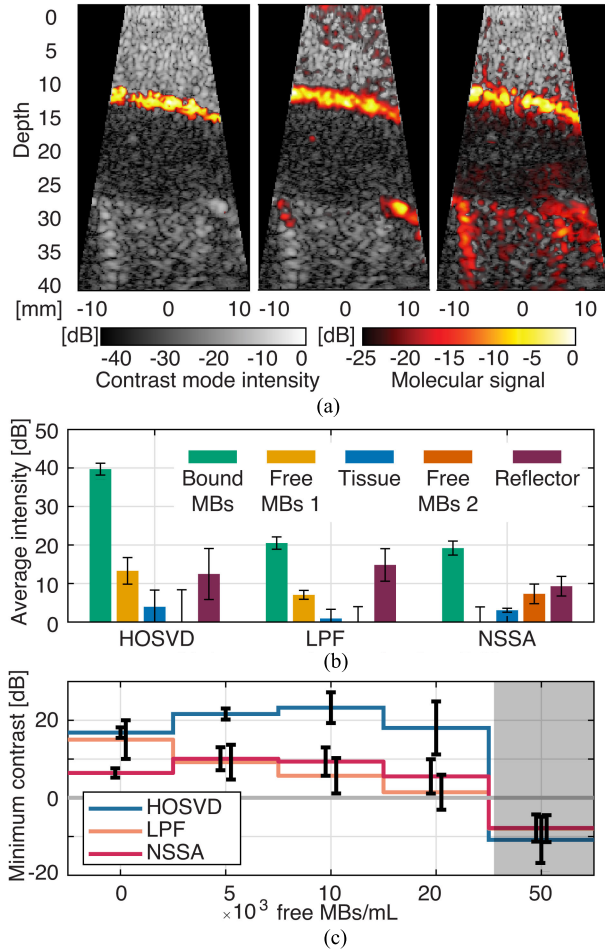


Fig. 9. Comparison of UMI techniques at 10×10^3 free MBs/mL. (a) Molecular signal of the phantoms with bound MBs obtained with the proposed HOSVD method (left), LPF (middle), and NSSA (right). (b) Average intensity in the ROIs. (c) Minimum bound MBs contrast as a function of free MB concentration.

reflector. These observations were consistent across all free MBs concentrations, except at 50×10^3 MBs/mL where the molecular signal of bound MBs was below the other ROIs for all the three UMI techniques (see Fig. S3).

The average intensity of each ROI at 10×10^3 free MBs/mL is shown in Fig. 9(b), normalized to the minimum intensity of

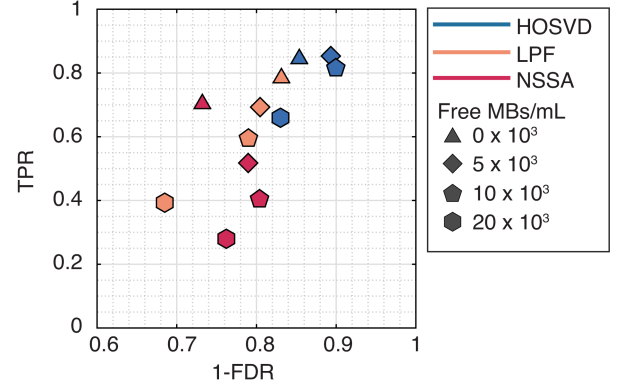


Fig. 10. TPR against FDR for the different UMI techniques. Higher TPR indicates more TPs, while higher 1-FDR indicates less FNs.

each technique. The intensity of bound MBs was much higher for HOSVD (40 dB) than for LPF (20 dB) and NSSA (19 dB). At this free MB concentration, the limiting ROIs were the free MBs 1 for HOSVD and the reflector for LPF and NSSA. The minimum contrast, defined as the intensity difference between bound MBs and limiting ROI [see Fig. 9(c)], was higher for HOSVD across all concentrations for which the filters worked, indicating a better performance. The average improvement across concentrations was 12.1 dB against both LPF and NSSA.

Finally, we studied the performance of the filters as binary classifiers, defining blocks above a given threshold as predicted positive (i.e., bound MBs expected) and predicted negative below the threshold (i.e., no bound MBs expected). Fig. 10 shows the TPR and false discovery rate (FDR) using an optimal threshold for concentrations up to 20×10^3 MBs/mL. At equal concentrations, HOSVD detected on average $1.34\times$ and $1.81\times$ more TPs compared to LPF and NSSA, and $1.73\times$ and $1.78\times$ less FPs, respectively.

V. DISCUSSION

In this study, we have applied HOSVD for UMI, eliminating the need for waiting steps and MB destruction, thus avoiding potential undesired biological effects [13]. HOSVD was used to decompose the signals within a block of pixels based on their temporal coherence and nonlinear content. We then separated bound MBs from freely circulating MBs and tissue by calculating a bound MB indicator χ , defined in (2), which increased in the presence of nonlinear signals that were also temporally coherent with tissue, i.e., signals from bound MBs [see Fig. 4(f)]. The validity of the concept was proven in silico [see Fig. 5(a)] and in vitro [see Fig. 6(g)]: higher values of the proposed parameter χ were observed for blocks containing bound MBs than those containing only tissue or free MBs. We then used χ to estimate the molecular signal (i.e., signal from bound MBs), filtering out blocks with low χ while retaining blocks with a high χ . The distribution of molecular signal measured with this method agreed with the molecular signal measured with DTE (see Fig. 7) and was able to detect the presence of bound MBs for concentrations of free MBs $\leq 20 \times 10^3$ MBs/mL. While the HOSVD technique failed at

50×10^3 MBs/mL, so did the other nondestructive techniques LPF and NSSA [see Fig. 9(c)]. At the same time, such a high concentration is only expected in vivo for a few seconds after a bolus injection [43]. On the other hand, HOSVD was the only technique that exclusively retained the molecular signal from bound MBs in the presence of $\leq 20 \times 10^3$ free MBs/mL (see Fig. 9) as LPF showed leakage from the first cloud of free MBs and the reflector, while NSSA showed leakage from the second cloud of free MBs and the reflector (see Fig. 9). As a result, HOSVD detected a larger fraction of TPs and lower number of FPs than both LPF and NSSA.

Notably, LPF and NSSA struggled to filter out the nonlinear signals from the embedded reflector whereas HOSVD did not (see Fig. 9). Likely, this signal was not generated by the reflector, but originated from the very efficient reflection of the nonlinear content of incoming wave. This explanation is supported by an increase in the intensity of the reflector in the AM acquisitions with increasing concentrations of free MBs [see Fig. 6(a)], as it is known that high numbers of MBs increase the nonlinear propagation in the imaging wave [23]. We attribute the better performance of HOSVD in the presence of nonlinear propagation to the definition of χ , which uses the fraction of nonlinear to total intensity and was higher for bound MBs than for the reflector.

Note that, in these experiments, we did not use ARF pulses for NSSA as in previous studies [28]. The NSSA method filters out both low NSSA values belonging to tissue, and high NSSA values belonging to free MBs, retaining mid-NSSA values of bound MBs. In previous studies, ARF pulses were used to increase the NSSA of bound MBs compared to tissue [24]. However, in our experiments, the NSSA of bound MBs overlapped with the NSSA of free MBs (see Fig. S4) rather than tissue. Therefore, we suspect that ARF pulses would have further increased the NSSA of bound MBs, worsening the overlap with free MBs. In addition, it is likely that ARF pulses would have induced nonspecific MB binding [27], thereby preventing an accurate comparison between techniques.

The HOSVD method relies on separating the blocks containing bound MBs owing to a higher χ value. The in vitro experiments showed a χ decrease with increasing concentrations of free MBs [see Fig. 6(c)], which effectively lowered the molecular signal intensity (see Fig. 8). We expect this trend to be due to two reasons. First, the blocks contained both bound and free MBs, so increasing the concentration of free MBs raised the energy of higher temporal ranks [44], thereby increasing the denominator in (2). Second, higher concentrations of free MBs lead to greater acoustic attenuation, consequently reducing the nonlinear backscattering from the bound MBs. This nonlinear backscattering is further attenuated on its way to the transducer, effectively decreasing the numerator in (2). In practice, higher transmitted pressures could partly compensate for the increased attenuation at high concentrations of MBs. Additionally, the nonlinear backscattering of bound MBs, and thus χ , could be increased by optimizing the MB formulation and/or the pulse sequence. As such, monodisperse MBs [45] with matched resonance to the imaging frequency would provide higher nonlinear content than polydisperse MBs [46]. Regarding the pulse

sequence, we tested a pulse inversion (PI) sequence of three pulses of equal amplitude but inverse polarity using the same experimental conditions and analysis pipeline (i.e., block-wise HOSVD). However, PI acquisitions yielded a lower χ for bound MBs (data not shown), resulting in a lower performance than AM. In the future, other sequences such as AM with multiple levels [47] or AM with PI [20] could be tested.

In order to filter out blocks without bound MBs, an adequate χ threshold was necessary. In our experiments, the χ values of the blocks containing bound MBs ranged from 0.10 to 0.16 in silico and from 0.1 to 0.4 in vitro. We expect χ to vary owing to multiple factors, such as the tissue intensity, amount of free MBs and bound MBs, and the MBs nonlinear backscattering coefficient. Therefore, in practice, a recording specific threshold could be defined using an acoustic acquisition following the injection of MBs, during which no significant binding is still expected. The χ distribution in this acquisition would indicate the appropriate threshold to retain only blocks with bound MBs.

In this study, we have introduced the pre-calculation of the average and nonlinear signals (see Fig. 1) before computing the HOSVD as a novelty [32]. This step reduced the tensor length in the pulsing dimension and, thus, the number of ranks along that dimension, from three to two. Without preprocessing, the second rank normally contains the difference between the pulses with odd and even elements (i.e., residual) [32]. However, at the intersection of two ROI (e.g., tissue + free MBs, or tissue + bound MBs), the residual and nonlinear content did not separate into unique ranks (see Fig. S5(a) and (b)). This rank mixing led to artifacts in the calculation of χ [see Fig. S5(c)]. Outside the artifact regions, the preprocessing did not influence the χ difference between bound MBs, tissue, and free MBs [see Fig. S5(c) and (d)].

DTE was used to qualitatively validate the HOSVD method (see Fig. 7), as the performance of DTE as a function of the free MBs concentration could not be assessed. Both methods detected some molecular signal from areas that were not in contact with MBs [see Fig. 7(a)]. These signals were likely caused by gaseous particles, as detection with DTE implies that the signal sources disappeared after the high-intensity bursts. Cross contamination with MBs could not account for the signal in the left bottom of the phantoms without bound MBs in Fig. 7(a) as they had not been in contact with MBs. Thus, we hypothesize that these signals belong to small gas pockets that form when the phantoms warmed to room temperature. In any case, these signals did not influence the validation of the technique, as they were excluded from the analyzed ROIs.

Our definition of χ poses two practical limitations. First, the signal of bound MBs is expected to group in the first temporal rank only in the absence of tissue motion. It is known that tissue motion will lead to a distribution of the signal over multiple temporal ranks, and likewise, the signals of bound MBs, that move with the tissue, will be distributed in an equal manner. In that case, (2) could be adjusted to include the nonlinear energy of those ranks in the numerator, rather than just the first rank. The performance of this adjusted approach should be explored in future studies. Second, our technique classifies

strong static nonlinear signals as molecular signals, relying on the hypothesis that bound MBs also behave nonlinearly, which seems appropriate based on previous literature [48], [49]. Therefore, slow nonbound MBs could contaminate the value of χ and lead to FPs if the acquisition time is not long enough to capture a temporal pattern that is strong enough to be decomposed into separate ranks. Furthermore, the current technique cannot distinguish between specific or unspecific bound MBs [50]. Future research could combine the HOSVD method with multifrequency pulse sequences to determine the binding specificity, as previous studies suggested that specific binding could be determined from the MB response at various frequencies [51], [52].

The experimental validation was limited to one imaging frequency (2 MHz), one speed of free MBs (~ 10 cm/s), and one concentration of bound MBs (~ 300 MBs/mm²). We expect that our proposed method will still work as long as the chosen center frequency induces a nonlinear response from the MBs, and free MBs move faster than tissue. The exact impact of variations in the previous parameters, the influence of tissue motion and the minimum detectable concentration of bound MBs should be investigated in future studies. Additionally, the influence of the pixel-block size should be addressed, as previous studies using block-wise SVD [24], [53] showed that various metrics improved by increasing the block size, up to an optimal size, at the expense of computational time. Importantly, since rapid imaging is a key advantage of this method against DTE, computational time should be kept low. Currently, the analysis of a single 1-s recording using MATLAB in 16 CPUs took ~ 2 s. Further algorithm optimization, downsampling or downsizing of the ROIs, and/or truncated HOSVD that discard irrelevant temporal ranks could further increase the speed to achieve real-time imaging.

The validation of our HOSVD method was limited to in vitro acquisitions, as an in vivo validation was beyond the scope of this study. We anticipate that the performance of HOSVD for in vivo acquisitions will depend on the specific application. Specifically, HOSVD will be influenced by factors such as the concentration of bound and free MBs, free-MB velocities, or tissue motion. The in vitro results presented here emulate fast free-MB velocities and low tissue motion, such as those in large-diameter vasculature. However, they are less similar to scenarios with slower free-MB velocities, such as those found in microvasculature. Nonetheless, we anticipate that the technique may enable the discrimination of bound MBs from slow-moving MBs, particularly if the acquisition length and/or the imaging frequency (and thus the spatial resolution) are increased. Additionally, we expect a bigger challenge for the use of the method in the presence of large tissue motion, for which further modifications such as motion compensation will likely be necessary. In any case, future in vivo validation will be essential to assess the method's performance across the aforementioned scenarios.

VI. CONCLUSION

Our results indicate that the ultrasound molecular signal obtained with the proposed HOSVD method was in agreement with the gold standard DTE, albeit that the HOSVD method

does not require destructive pulses and can be used in near real time. We also showed that HOSVD separated the bound MB signal from tissue and $\leq 20 \times 10^3$ free MBs/mL consistently better than LPF and NSSA. This improved separation allowed HOSVD to detect, simultaneously, a larger fraction of TPs and a lower number of FPs. These findings show the potential of HOSVD for rapid, nondestructive UMI.

APPENDIX

While in the main text, we focused on HOSVD given its better performance in various applications [30], [32], conventional SVD may also be used for this purpose. In this section, we briefly compare the performance of both techniques for our specific application.

For this comparison, we used the same in vitro recordings as in the results section. Similar to the HOSVD approach, we applied block-wise SVD to 5×5 pixel blocks with a 20-pixel overlap. Since conventional SVD is limited to 2-D decomposition, we first concatenated the average and nonlinear signals along the spatial dimension, forming a $(2N_x N_z) \times N_t$ Casorati matrix. In this matrix, the average signals occupy the first $N_x N_z$ rows, while the nonlinear signals are positioned from $N_x N_z + 1$ to $2N_x N_z$. This concatenation into a single matrix ensures a shared temporal decomposition for both the average and nonlinear signals, which is essential for the proposed method.

The SVD of the Casorati matrix reads

$$T = USV^* \quad (a1)$$

where $*$ denotes the conjugate transpose. Based on the SVD components, we then define an analogous bound MB indicator

$$\chi_{\text{SVD}} = \frac{\sqrt{\sum_{i=N_x N_z + 1}^{2N_x N_z} \sum_j^{N_t} (U_{i,1} S_{1,1} V_{1,j}^*) (U_{i,1} S_{1,1} V_{1,j}^*)^*}}{\sqrt{\sum_{i=1}^{2N_x N_z} \sum_j^{N_t} (U_{i,1} S_{1,1} V_{1,j}^*) (U_{i,1} S_{1,1} V_{1,j}^*)^*}} \quad (a2)$$

We used the bound MB indicator to compute the molecular signal as

$$s_{\text{SVD}} = s_{\text{AM}} \cdot \Phi((\chi - \mu_{\text{SVD}})/\sigma_{\text{SVD}}) \quad (a3)$$

where μ_{SVD} and σ_{SVD} were optimized for HOSVD.

Fig. 11 compares the molecular signal without free MBs for the SVD and HOSVD methods. The bound MBs at the surface were also detected by means of conventional SVD, yet increased leakage from the moving particles in front of the phantom was observed compared to the HOSVD. Motion between pulses of the AM sequence may introduce a nonlinear signal artifact [54], and the increased leakage suggests that SVD was more affected by this artifact than HOSVD.

The molecular signal for the other concentrations is shown in Fig. S6. Similar to HOSVD, conventional SVD was able to detect the presence of bound MBs up to a concentration of 20×10^3 free MBs/mL, although the total area of molecular signal was lower. This resulted effectively in a lower TPR

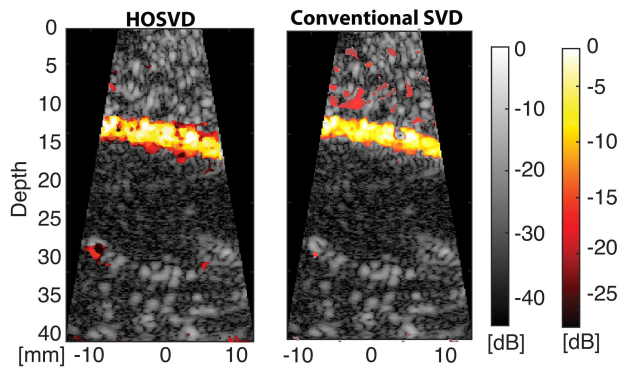


Fig. 11. Molecular signal (in yellow-red) overlaid on an AM frame (gray) obtained using HOSVD (left) and conventional SVD (right) in the acquisition with bound MBs and no free MBs.

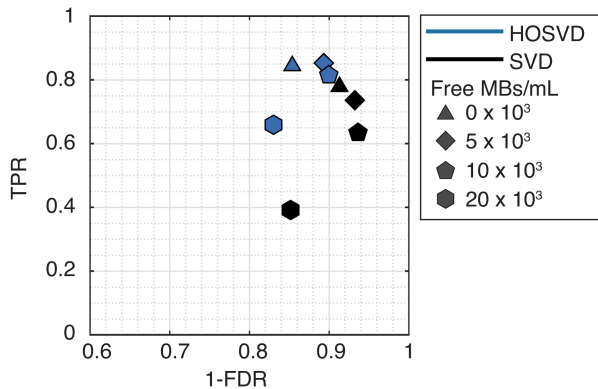


Fig. 12. TPR against FDR for HOSVD and conventional SVD.

for conventional SVD (see Fig. 12). The reduced TPR can be attributed to the less effective filtering of the interpulse motion artifact. Raising the detection threshold to improve the TPR would in turn increase the fraction of FNs in the free MBs ROIs. It is worth mentioning that SVD had a lower false discovery rate (FDR) than HOSVD. Still, a similar FDR could be achieved with HOSVD by decreasing the detection threshold, making the technique more robust to FPs at the expense of reducing the sensitivity to TPs.

These findings illustrate that the proposed principle can be applied using conventional SVD, which may offer a reduced computational load than HOSVD at the expense of lower sensitivity to bound MB.

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value decomposition (SVD) in the context of their study, which forms the basis of the results presented in the Appendix.

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